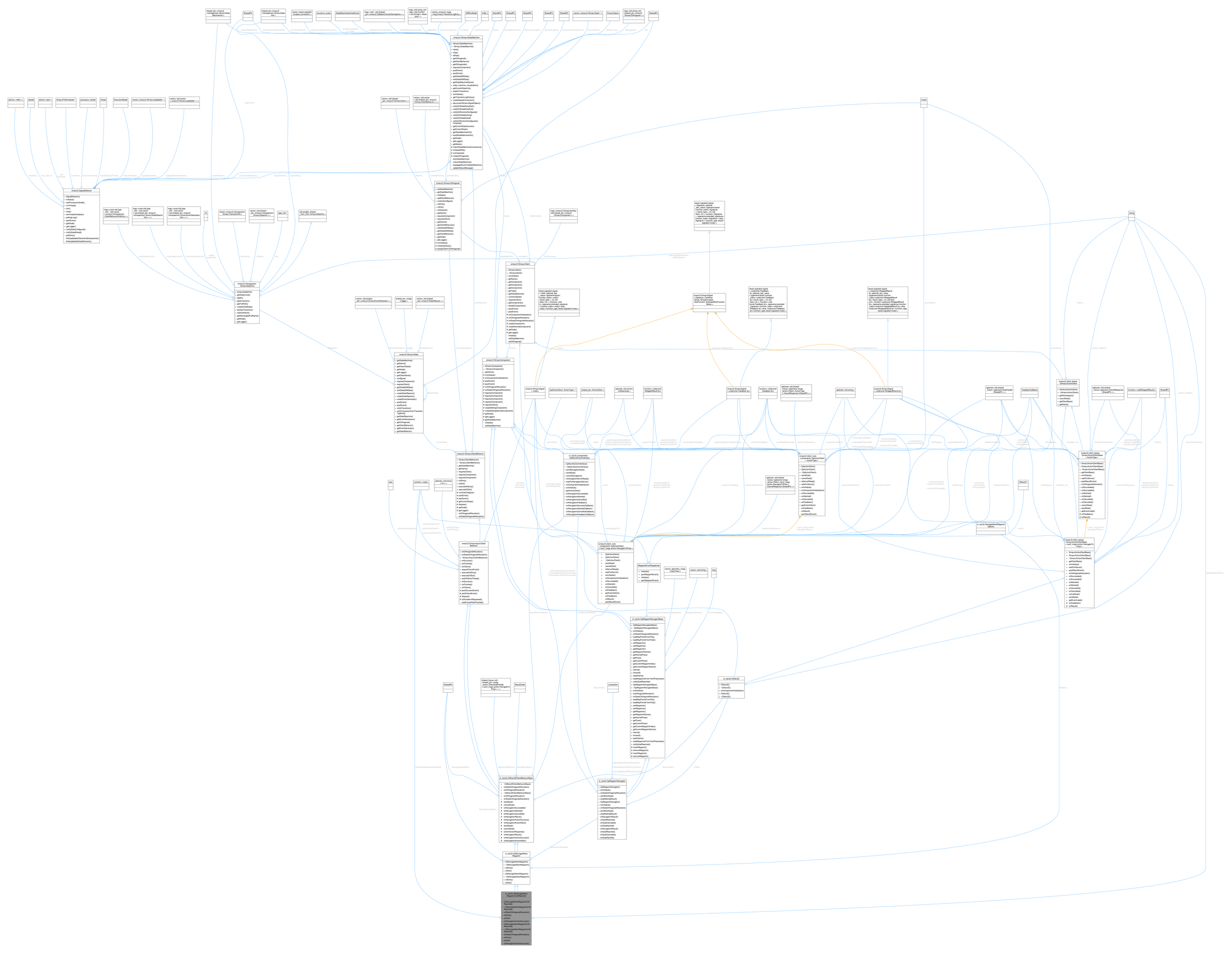

shared_ptr< smacc2::introspection::SmaccStateMachineInfo >
SharedPtr
shared_ptr< smacc2::introspection::SmaccStateInfo >
vector< boost::signals2::scoped_connection >
recursive_mutex
StateMachineInternalAction
map< void *, std::shared_ptr< smacc2::CallbackCounterSemaphore > >
map< std::string, std::pair< std::function< std::string()>, boost::any > >
vector< smacc2_msgs::msg::SmaccTransitionLogEntry >
SMRunMode
int64_t
SharedPtr
SharedPtr
SharedPtr
SharedPtr
SharedPtr
vector< smacc2::ISmaccState * >
SmaccStatus
map< std::string, std::shared_ptr< smacc2::ISmaccOrthogonal > >
SharedPtr
#stateMachineInfo_
#stateMachineStatusPub_
#currentStateInfo_
#longLivedSignalConnections_
-eventQueueMutex_
-m_mutex_
-stateMachineCurrentAction
-stateCallbackConnections
-globalData_
-transitionLogHistory_
-runMode_
-stateSeqCounter_
#timer_
#stateMachinePub_
#transitionLogPub_
#eventsLogPub_
#transitionHistoryService_
#currentState_
#status_msg_
#orthogonals_
#nh_
-statusPub_
smacc2::ISmaccStateMachine
+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExitting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfiguration Finished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()
atomic< int64_t >
double
atomic< bool >
SmaccFifoScheduler *
processor_handle
thread
ExecutionModel
vector< smacc2::ISmaccUpdatable * >
vector< std::vector< smacc2::ISmaccUpdatable * > >
-lastState_
-loop_rate_hz
-end_
-initialized_
-scheduler_
-processorHandle_
-signalDetectorThread_
-executionModel_
-updatableClients_
-updatableStateElements_
+stateMachine_
-smaccStateMachine_
-signalDetector_
+parentState_
smacc2::SignalDetector
+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ pollingLoop()
+ postEvent()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()
map< const std::type_info *, std::vector< smacc2::introspection::ClientBehaviorInfoEntry > >
map< const std::type_info *, std::vector< std::shared_ptr< smacc2::introspection::SmaccStateReactorInfo > > >
map< const std::type_info *, std::vector< std::shared_ptr< smacc2::introspection::SmaccEventGeneratorInfo > > >
int
vector< smacc2::introspection::SmaccTransitionInfo >
vector< std::shared_ptr< smacc2::introspection::SmaccStateInfo > >
type_info *
std::enable_shared_from_this< SmaccStateInfo >
+staticBehaviorInfo
+stateReactorsInfo
+eventGeneratorsInfo
+depth_
+stateIndex_
+transitions_
+children_
+tid_
smacc2::introspection::SmaccStateInfo
+ SmaccStateInfo()
+ getStateLevel()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()
#stateInfo_
vector< std::shared_ptr< smacc2::ISmaccClient > >
vector< std::vector< std::shared_ptr< smacc2::ISmaccClientBehavior > > >
#clients_
-clientBehaviors_
-stateMachine_
-mutex_
smacc2::ISmaccOrthogonal
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()
map< smacc2::ComponentKey, std::shared_ptr< smacc2::ISmaccComponent > >
#node_
-stateMachine_
#stateMachine_
-orthogonal_
#components_
+demangledStateName
+fullStateName
smacc2::ISmaccClient
+ ISmaccClient()
+ ~ISmaccClient()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
# onComponentInitialization()
# onOrthogonalAllocation()
# onStateOrthogonalAllocation()
# createComponent()
# createNamedComponent()
# getNode()
# getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()
boost::signals2::signal < Signature, optional _last_value< typename boost ::function_traits< Signature >::result_type >, int, std ::less< int >, function< Signature >, typename extended_signature< function_traits< Signature >::arity, Signature >::function_type, boost:: signals2::mutex >
boost::signals2::signal < void(), optional_last _value< typename boost:: function_traits< void()> ::result_type >, int, std ::less< int >, function< void ()>, typename extended_signature < function_traits< void()>::arity, void()>::function_type, boost::signals2::mutex >
smacc2::SmaccSignal < Signature, Combiner, Group, GroupCompare, SlotFunction, ExtendedSlotFunction, Mutex >
boost::signals2::signal < void(const Feedback &), optional_last_value < typename boost::function _traits< void(const Feedback &)>::result_type >, int, std ::less< int >, function< void (const Feedback &)>, typename extended _signature< function_traits< void(const Feedback &)>::arity, void(const Feedback &)>::function_type, boost::signals2::mutex >
boost::signals2::signal < void(const WrappedResult &), optional_last_value <typename boost::function _traits<void(const WrappedResult &)>::result_type>, int, std::less <int>, function<void(const WrappedResult &)>, typename extended_signature<function _traits<void(const WrappedResult &)>::arity, void(const WrappedResult &)>::function_type, boost::signals2::mutex >
vector< std::shared_ptr< smacc2::SmaccEventGenerator > >
shared_ptr< rclcpp ::Logger >
vector< std::shared_ptr< smacc2::StateReactor > >
#eventGenerators_
#logger_
#stateReactors_
smacc2::ISmaccState
+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransition TypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()
#parentState_
-currentOrthogonal
-stateMachine_
#owner_
smacc2::ISmaccComponent
+ ISmaccComponent()
+ ~ISmaccComponent()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onOrthogonalAllocation()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()
< void()>
< void(const Feedback &)>
< void(const WrappedResult &)>
mutex
string
-actionMutex_
#name_
+action_endpoint_
smacc2::SmaccSignal < void()>
CpActionClient< ActionType > *
shared_ptr< ActionClient >
optional< std::chrono ::milliseconds >
function< void(const WrappedResult &)>
smacc2::SmaccSignal < void(const Feedback &)>
function< void(const Feedback &)>
optional< std::shared _future< typename rclcpp _action::Client< ActionType >::CancelResponse::SharedPtr > >
optional< std::string >
smacc2::SmaccSignal < void(const WrappedResult &)>
optional< std::shared _future< typename GoalHandle ::SharedPtr > >
FeedbackCallback
smacc2::client_bases ::ISmaccActionClient
+ ISmaccActionClient()
+ ~ISmaccActionClient()
+ getNamespace()
+ cancelGoal()
+ getClientBase()
+ getName()
optional< std::shared _future< typename CancelResponse ::SharedPtr > >
function< void(WrappedResult)>
SharedPtr
smacc2::ISmaccClientBehavior
+ ISmaccClientBehavior()
+ ~ISmaccClientBehavior()
+ getStateMachine()
+ getName()
+ requiresClient()
+ requiresComponent()
+ requiresComponent()
+ onEntry()
+ onExit()
+ executeOnEntry()
+ executeOnExit()
# runtimeConfigure()
# postEvent()
# postEvent()
# getCurrentState()
# dispose()
# getNode()
# getLogger()
- onOrthogonalAllocation()
- onStateOrthogonalAllocation()
-currentState
-actionClient_
+onGoalAccepted_
+onGoalRejected_
-client_
+postNavigationAbortedEvent
+postNavigationCancelledEvent
+postNavigationSuccessEvent
+serverTimeout
+postAbortedEvent
+postCancelledEvent
+postSuccessEvent
+onNavigationFeedback_
+postNavigationFeedbackEvent
+onActionFeedback_
+postFeedbackEvent
-lastCancelResponse_
+actionServerName
+onNavigationAborted
+onNavigationCancelled
+onNavigationSucceeded
+onActionAborted_
+onActionCancelled
+onActionSucceeded_
+controllerName_
+goalCheckerName_
-lastRequest_
-feedbackCallback_
+onAborted_
+onCancelled_
+onSucceeded_
+lastRequest_
+feedback_cb
+lastCancelResponse_
#client_
-actionServerName_
cl_nav2z::components ::CpNav2ActionInterface
+ CpNav2ActionInterface()
+ ~CpNav2ActionInterface()
+ sendNavigationGoal()
+ sendGoal()
+ cancelNavigation()
+ isNavigationServerReady()
+ waitForNavigationServer()
+ onComponentInitialization()
+ onInitialize()
+ getActionClient()
+ onNavigationSucceeded()
+ onNavigationAborted()
+ onNavigationCancelled()
+ onNavigationFeedback()
- onNavigationSuccessCallback()
- onNavigationAbortedCallback()
- onNavigationCancelledCallback()
- onNavigationFeedbackCallback()
smacc2::client_core _components::CpActionClient < ActionType >
+ CpActionClient()
+ CpActionClient()
+ ~CpActionClient()
+ sendGoal()
+ cancelGoal()
+ isServerReady()
+ waitForServer()
+ onInitialize()
+ onComponentInitialization()
+ onSucceeded()
+ onAborted()
+ onCancelled()
+ onFeedback()
+ getActionClient()
- onFeedback()
- onResult()
- postResultEvent()
optional< std::shared _future< typename rclcpp _action::Client< nav2_msgs ::action::NavigateToPose > ::CancelResponse::SharedPtr > >
smacc2::client_bases ::SmaccActionClientBase < ActionType >
+ SmaccActionClientBase()
+ SmaccActionClientBase()
+ ~SmaccActionClientBase()
+ getClientBase()
+ onInitialize()
+ waitForServer()
+ postResultEvent()
+ onOrthogonalAllocation()
+ onSucceeded()
+ onSucceeded()
+ onAborted()
+ onAborted()
+ onCancelled()
+ onCancelled()
+ cancelGoal()
+ sendGoal()
+ getEventLabel()
# onFeedback()
# onResult()
ClNav2Z *
bool
function< void()>
optional< std::future < int > >
-isShutdownRequested_
-postFailureEventFn
-postFinishEventFn
-postSuccessEventFn_
-onEntryThread_
-onExitThread_
-onFailure_
-onFinished_
-onSuccess_
+postWaypointFn
-lastCancelResponse_
< nav2_msgs::action ::NavigateToPose >
+client_
smacc2::SmaccAsyncClient Behavior
+ onOrthogonalAllocation()
+ onStateOrthogonalAllocation()
+ ~SmaccAsyncClientBehavior()
+ onSuccess()
+ onFinished()
+ onFailure()
+ requestForceFinish()
+ executeOnEntry()
+ executeOnExit()
+ waitOnEntryThread()
+ onSuccess()
+ onFinished()
+ onFailure()
# postSuccessEvent()
# postFailureEvent()
# dispose()
# isShutdownRequested()
- waitFutureIfNotFinished()
smacc2::client_core _components::CpActionClient < nav2_msgs::action::NavigateToPose >
+ CpActionClient()
+ CpActionClient()
+ ~CpActionClient()
+ sendGoal()
+ cancelGoal()
+ isServerReady()
+ waitForServer()
+ onInitialize()
+ onComponentInitialization()
+ onSucceeded()
+ onAborted()
+ onCancelled()
+ onFeedback()
+ getActionClient()
- onFeedback()
- onResult()
- postResultEvent()
WaypointEventDispatcher
+ initialize()
+ postWaypointEvent()
+ initialize()
+ postWaypointEvent()
vector< geometry_msgs ::msg::Pose >
vector< std::string >
long
cl_nav2z::NavigateNextWaypoint Options
smacc2::client_bases ::SmaccActionClientBase < nav2_msgs::action::NavigateTo Pose >
+ SmaccActionClientBase()
+ SmaccActionClientBase()
+ ~SmaccActionClientBase()
+ getClientBase()
+ onInitialize()
+ waitForServer()
+ postResultEvent()
+ onOrthogonalAllocation()
+ onSucceeded()
+ onSucceeded()
+ onAborted()
+ onAborted()
+ onCancelled()
+ onCancelled()
+ cancelGoal()
+ sendGoal()
+ getEventLabel()
# onFeedback()
# onResult()
+onNavigationRequestAborted
+onNavigationRequestCancelled
+onNavigationRequestSucceded
#nav2ActionInterface_
+nav2ActionInterface_
-waypointsEventDispatcher
#waypoints_
#waypointsNames_
-currentWaypoint_
cl_nav2z::CpWaypointNavigatorBase
+ CpWaypointNavigatorBase()
+ ~CpWaypointNavigatorBase()
+ onInitialize()
+ onStateOrthogonalAllocation()
+ loadWayPointsFromFile()
+ loadWayPointsFromFile2()
+ setWaypoints()
+ setWaypoints()
+ getWaypoints()
+ getWaypointNames()
+ getNamedPose()
+ getPose()
+ getCurrentPose()
+ getCurrentWaypointIndex()
+ getCurrentWaypointName()
+ rewind()
+ forward()
+ seekName()
+ loadWaypointsFromYamlParameter()
+ notifyGoalReached()
+ CpWaypointNavigatorBase()
+ ~CpWaypointNavigatorBase()
+ onInitialize()
+ onOrthogonalAllocation()
+ onStateOrthogonalAllocation()
+ loadWayPointsFromFile()
+ loadWayPointsFromFile2()
+ setWaypoints()
+ setWaypoints()
+ getWaypoints()
+ getWaypointNames()
+ getNamedPose()
+ getPose()
+ getCurrentPose()
+ getCurrentWaypointIndex()
+ getCurrentWaypointName()
+ rewind()
+ forward()
+ seekName()
+ loadWaypointsFromYamlParameter()
+ notifyGoalReached()
# insertWaypoint()
# removeWaypoint()
# insertWaypoint()
# removeWaypoint()
cl_nav2z::ClNav2Z
+ ClNav2Z()
+ ~ClNav2Z()
+ onComponentInitialization()
+ ClNav2Z()
+ ~ClNav2Z()
SharedPtr
shared_future< std ::shared_ptr< rclcpp _action::ClientGoalHandle < nav2_msgs::action::NavigateTo Pose > > >
ResultCode
connection
#actionClient_
-postEvGoalWaypointReached_
#navigationCallback_
#goalHandleFuture_
#navigationResult_
-abortedNav2ZClientConnection
-cancelledNav2ZClientConnection
-succeddedNav2ZClientConnection
#nav2zClient_
-client_
#navigationCallback_
cl_nav2z::CbNav2ZClientBehaviorBase
+ ~CbNav2ZClientBehaviorBase()
+ onStateOrthogonalAllocation()
+ onOrthogonalAllocation()
+ ~CbNav2ZClientBehaviorBase()
+ onOrthogonalAllocation()
+ onStateOrthogonalAllocation()
# sendGoal()
# cancelGoal()
# onNavigationSucceeded()
# onNavigationAborted()
# onNavigationCancelled()
# onNavigationResult()
# onNavigationActionSuccess()
# onNavigationActionAbort()
# sendGoal()
# cancelGoal()
# isOwnActionResponse()
# onNavigationResult()
# onNavigationActionSuccess()
# onNavigationActionAbort()
cl_nav2z::CpWaypointNavigator
+ CpWaypointNavigator()
+ onInitialize()
+ onStateOrthogonalAllocation()
+ sendNextGoal()
+ stopWaitingResult()
+ CpWaypointNavigator()
+ onInitialize()
+ onStateOrthogonalAllocation()
+ sendNextGoal()
+ stopWaitingResult()
- onNavigationResult()
- onGoalReached()
- onGoalCancelled()
- onGoalAborted()
- onNavigationResult()
- onGoalReached()
- onGoalCancelled()
- onGoalAborted()
#waypointsNavigator_
#options_
-goalWaypointName_
cl_nav2z::CbNavigateNext Waypoint
+ CbNavigateNextWaypoint()
+ ~CbNavigateNextWaypoint()
+ onEntry()
+ onExit()
+ CbNavigateNextWaypoint()
+ ~CbNavigateNextWaypoint()
+ onEntry()
+ onExit()
cl_nav2z::CbNavigateNext WaypointUntilReached
+ CbNavigateNextWaypointUntil Reached()
+ ~CbNavigateNextWaypointUntil Reached()
+ onStateOrthogonalAllocation()
+ onEntry()
+ onExit()
+ onNavigationActionSuccess()
+ CbNavigateNextWaypointUntil Reached()
+ ~CbNavigateNextWaypointUntil Reached()
+ onStateOrthogonalAllocation()
+ onEntry()
+ onExit()
+ onNavigationActionSuccess()